

Arduino Calculator

Conception phase – Microcontroller calculator using RS232

The project involves designing and implementing a simple calculator that communicates between a microcontroller and a computer program via a serial RS232 connection. The computer sends mathematical expressions to the microcontroller via user input, e.g. 12×47 or $15 \div 4$. The microcontroller receives and evaluates the expression, calculates and sends the result back to the computer. The computer then outputs the result and logs all transactions to a log file in text format.

User Interface

The operator will interact with the computer by entering expressions in a console. When the console is launched. The operator must select which COM port the microcontroller/ Arduino (FT232 module) is connected to. The baudrate, parity bits and stop bits can be optionally set in case the serial interface is set to a default. After a serial connection is established, the user can enter mathematical expressions via the console.

The computer then transmits the expression to the Arduino as a ASCII string through the FT232 module. The Arduino checks the syntax, parses the expression, calculates the result, and sends a response string back, e.g. Answer = 1279, or Error: Cannot divide by zero.

The computer displays the string in the console and logs the transactions to a log file. The Arduino will show successful communication by facilitating two LED's showing error(red) or communication active(green), and a LCD display showing status of current task – showing successful communication between PC and microcontroller, current expression, and result.

Fundamental functionalities

PC Console

- Setup and configure the serial connection (COM port, baudrate, parity bits, stop bits).
- Indicate successful connection with Arduino
- Error handling – disconnection, invalid expression, etc.
- Enter mathematical expression in the console.
- Indicate that the Arduino has received the input.
- Display the result from Arduino in the console window.
- Log all sent and received message in a text file.

Module Structures

PC Console

- Communication module – ensuring successful communication between Arduino and program.
- Error module – capturing all errors, from communication to invalid syntax.

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- Logging module – logging all events to a log file.
- Interface module – console program, accepting user input, displaying responses from microcontroller

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- Communication module – ensuring successful communication from PC.
- Display module – LED's showing error and LCD screen displaying status.
- Parser module – splitting expressions into arguments - argument 1, (math expression) argument 2, etc. Also checks syntax.
- Math module – does the arithmetic - (+, -, ÷, x), also checks for division by zero.
- Results module – formats output and sends back to console.

Programming language – the PC console

I have chosen C++ as the programming language, mainly because:

- Strings are handled easier (std: string) compared to the C language.
- C++ makes it easier to separate modules with classes.
- C requires more work when checking for errors.
- C++ is easier to read and maintain.
- C++ is used more commonly for PC applications as it has more support for larger programs.